CASE STUDY

Book Blog Generator Agent

End-to-End AI Content Automation with Substack Integration

Project Scope

An intelligent AI-powered system that automates the entire content creation pipeline for a book review blog. The agent discovers relevant books, generates high-quality blog posts matching the author's unique voice, and publishes directly to Substack as drafts—transforming what was a multi-hour manual process into a streamlined, automated workflow.

The Problem

Content creators face significant challenges maintaining a consistent publishing schedule:

- **Time-intensive research:** Finding relevant books in specific niches requires hours of searching and evaluation
- **Writer's block and consistency:** Maintaining a consistent voice and quality across dozens of posts is mentally exhausting
- **Manual publishing workflow:** Copying content to Substack, formatting, and setting up drafts adds friction to every post
- **Tracking overhead:** Remembering which books have already been covered to avoid duplicates

- **Scaling limitations:** Growing content output means proportionally more time investment

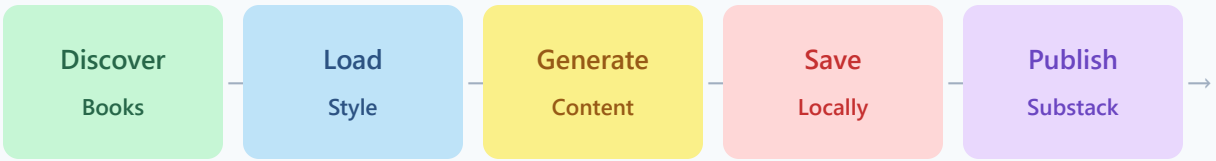
The Solution

A fully autonomous AI agent that handles every step of the content pipeline:

- **Intelligent discovery:** Automatically finds marketing, copywriting, and business books via Google Books API or curated lists
- **Style learning:** Analyzes example posts to match the author's exact tone, structure, and format
- **Quality content generation:** Produces publication-ready blog posts with actionable insights
- **Direct Substack publishing:** Browser automation creates drafts in Substack, ready for review
- **Automatic tracking:** Maintains records of completed books to prevent duplicates

System Architecture

Six-Stage Automation Pipeline



Track

Complete

AI Agent with Tool-Based Architecture

Built on OpenAI's GPT-4o with function calling, the agent operates through seven specialized tools:

- **search_books** – Query Google Books API
- **get_curated_books** – Load pre-selected book list
- **get_written_books** – Check completion history
- **load_style_examples** – Analyze writing samples
- **save_blog_post** – Write Markdown files
- **mark_book_complete** – Update tracking
- **publish_to_substack** – Automate publishing

Substack Integration

The system features deep integration with Substack through browser automation powered by Playwright:

- **Persistent authentication:** Uses a dedicated browser profile that maintains login sessions across runs
- **Smart content handling:** Automatically converts Markdown to Substack's editor format, applying proper heading styles
- **Book cover images:** Downloads cover images from Google Books, resizes for optimal display, and pastes via clipboard

- **Draft creation:** Posts are saved as drafts, allowing human review before publishing
- **Auto-save detection:** Waits for Substack's auto-save to confirm successful draft creation

Key Features

Multiple Operation Modes

Auto-discover, manual entry, list preview, and interactive chat modes for flexibility

Style Matching

Learns from example posts to replicate exact tone, structure, and formatting

Duplicate Prevention

Tracks all written books to ensure fresh content every time

Conversational Interface

Chat naturally with the agent to request specific books or customize output

Results & Impact

15+
Blog Posts Generated

90%
Time Saved Per Post

100%
Automated Pipeline

- Reduced blog post creation from 2-3 hours to under 15 minutes of review time
- Maintained consistent voice and quality across all generated content
- Eliminated manual copy-paste workflow with direct Substack publishing
- Enabled scaling content output without proportional time investment

- Created a reusable framework adaptable to other content niches

Technology Stack

Python | OpenAI GPT-4o | Function Calling | Playwright | Google Books API | Browser Automation | Substack API | Markdown | PIL/Pillow | CLI Interface

Conclusion

This project demonstrates expertise in building practical AI agents with tool-based architectures, external API integrations, and browser automation. It showcases the ability to create autonomous systems that solve real content marketing challenges while maintaining brand voice consistency and seamlessly integrating with existing publishing platforms like Substack.